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Post-Assembled Alkylphosphonic Acids for Efficient and Stable Inverted Perovskite Solar Cells

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The self-assembled monolayer (SAM) is now widely applied at the hole-selective interface of efficient inverted perovskite solar cells (PSCs). However, voids are unavoidable in SAMs due to the rough substrate and the deposition conditions, which lead to direct contact between the active layer and electrode, undermining the efficiency and stability of PSCs. Here, alkylphosphonic acids with different alkyl chain length are post-assembled to fill the voids, thereby forming a compact po-SAM layer. By further modifying the head group of alkylphosphonic acids, the efficiency and stability of PSCs are improved due to wetting and passivation. Finally, the po-SAM layer formed by (2-Bromoethyl) phosphonic acid and [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid contributes to a champion device with the power conversion efficiency of 25.16% (certified 24.67%) and improved stability under operation or storage. This work demonstrates the advantages of po-SAM layer in improving the performance of PSCs, offering ideas for SAM modification.

1. Introduction

Inverted perovskite solar cells (PSCs) have attracted great attention due to their advantages of stability,[1–3] low hysteresis,[4–6] and simple fabrication process. Recently, the power conversion efficiency (PCE) of inverted PSCs experienced rapid development, approaching that of their regular counterparts.[7–9] Apart from the progress in perovskite crystallization and defects passivation, the application of the self-assembled monolayer (SAM) at the hole-selective interface of inverted PSCs has made a significant contribution.[10,11,12] The SAM molecules usually consist of three parts, including an anchor group that can bind with the substrates, a spacer to facilitate free rotation for thermodynamically stable self-assembly, and a functional head group to realize efficient charge extraction.[13–15] Benefiting from the rational design, the SAM molecules with low optical and electrical loss such as [2-(9H-Carbazol-9-yl)ethyl]phosphonic acid, [2-(3,6-Dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid and [4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) stand out.[16–18] Among them, Me-4PACz with suitable dipole usually results in higher open-circuit voltage and was widely used.[19,20]

However, SAM inevitably has voids due to the deposition conditions and the morphology of the substrates,[21,22] especially the rough fluorine-doped tin oxide-coated glass (FTO) substrate with a textured surface that is beneficial for eliminating the coherence of the incoming light and affords light trapping by increasing the optical path length.[23,24] The voids in SAM will provide channels for the direct contact between the active perovskite layer and the electrode, leading to severe non-radiative charge recombination and threatening the operation stability of the devices.[25–27] In previous works, a co-assembly strategy was proposed by mixing a major SAM component and a second SAM component. Getautis et al. mixed a butylphosphonic acid with V1036 to form a co-SAM with improved coverage.[10] To further improve the quality of co-SAM, the phosphonic anchor group of the second SAM component was replaced by a carboxylic group with weaker acidity to ensure the robust anchoring of the major SAM component.[28,29] However, weaker acidity usually implies weaker binding with the substrate, which may compromise the stability of the devices.[30,31]

To improve the coverage of SAM without affecting the anchoring of the major SAM component nor lowering the binding strength between the second SAM component and the substrate, a post-assembly strategy is proposed here: the major SAM component is firstly deposited to anchor with the substrate, getting rid of the competition with the second component. Subsequently, the second component with the strong phosphonic anchor group can be deposited to fill the voids and form a compact and stable po-SAM.[10,31] In this work, a series of alkylphosphonic acids with different alkyl chain lengths are post-assembled to fill the voids of Me-4PACz and it is found that ethanephosphonic acid (EPA) with the shortest alkyl chain length performs best in improving the SAM coverage due to the smallest steric hindrance.[32] By further endowing EPA with a functional head group, the po-SAM layer formed by (2-Bromoethyl) phosphonic acid (Br-EPA) and Me-4PACz improves the wettability for perovskite, manipulates

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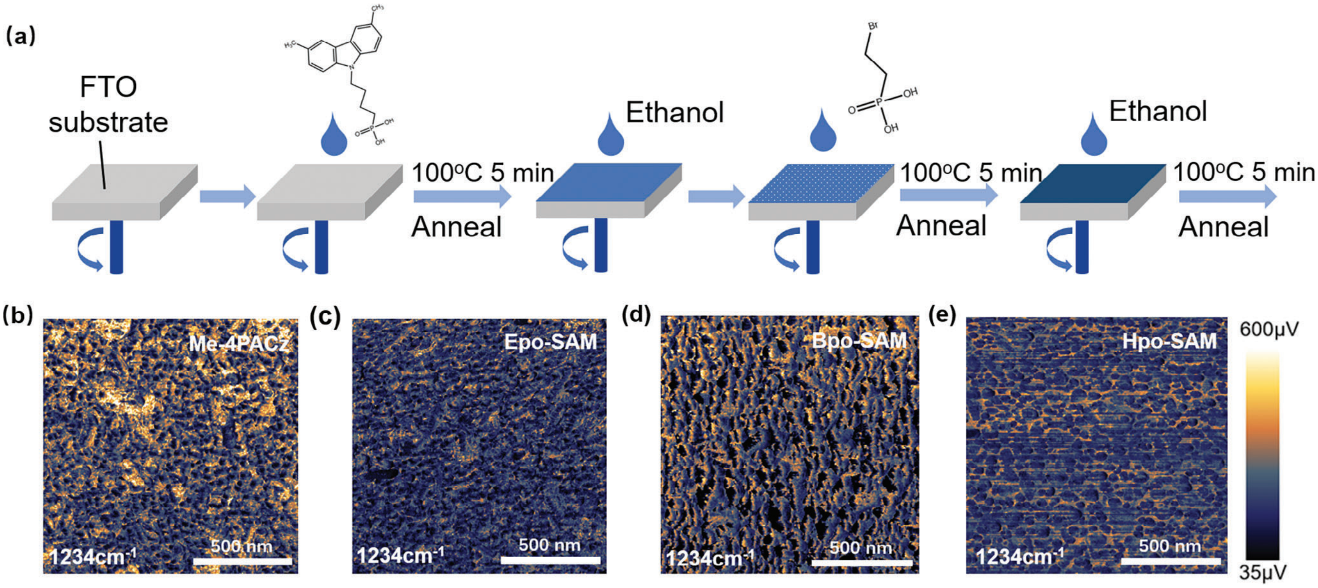

Figure 1. a) The schematic process of po-SAM deposition. b–e) Photo-induced surface potential mapping at wavenumber of 1234 cm^{-1} for FTO/Me-4PACz, FTO/Epo-SAM, FTO/Bpo-SAM, and FTO/Hpo-SAM, respectively. The concentration of the alkylphosphonic acids is 0.2 mg mL^{-1} for all the po-SAM samples.

the crystallization of the perovskite and passivates the defects at the buried interface. Finally, a champion device with FTO/po-SAM achieved a PCE of 25.16% (certified PCEs of 24.67% and 24.61% under forward and reverse scan, respectively, with an aperture area of 0.0817 cm^2). The PSCs were stable, maintaining >90% of the initial PCE after 1000 h of operation at MPP, under 1-sun illumination, 40 °C. This work offers a new and easy way to deposit a compact and functional SAM for further improving the performance of PSCs.

2. Results and Discussion

The schematic process of depositing po-SAM is shown in **Figure** 1a and the molecular structures of Me-4PACz and alkylphosphonic acids with different alkyl chain length (ethanephosphonic acid, EPA; butylphosphonic acid, BPA; hexylphosphonic acid, HPA) are shown in Figure S1 (Supporting Information). The solvent rinse step before and after the deposition of alkylphosphonic acids is to wash away the redundant Me-4PACz and alkylphosphonic acids, respectively. The first-principles density functional theory (DFT) calculations were performed to investigate the po-SAM deposition process on FTO surfaces, which is based on an ideal plane adsorption model (Figure S2, Supporting Information). As a result, the adsorption energy for the assembly of Me-4PACz and the post-assembly of EPA are −1.09 and −1.12 eV, respectively. The negative values of adsorption energy indicate that these adsorption processes are thermodynamically favorable.

To initially determine the concentration of alkylphosphonic acids, the devices were fabricated with the structure of FTO/SAM/perovskite/phenyl-C_{61}-butyric acid methyl ester (PCBM)/ bathocuproine (BCP)/Ag. For each batch, twelve devices were fabricated and the devices' performance are summarized in Figure S3 (Supporting Information). It is found that the devices' performance is improved for all the devices with the increase in the concentration of alkylphosphonic acids and almost keeps at a constant value after the concentration reaches 0.2 mg mL^{-1}, indicating that the amount of alkylphosphonic acids is adequate at this stage for filling most of the voids in Me-4PACz layer and the redundant molecules could be washed away by the subsequent solvent rinse step. The detailed photovoltaic parameters for the devices using 0.2 mg mL^{-1} alkylphosphonic acids are shown in Figure S4 (Supporting Information). As a result, the devices based on the po-SAM formed by Me-4PACz and EPA (Epo-SAM) show better PCE (22.48 ± 0.37%) than the devices based on Me-4PACz (21.13 ± 0.75%), Bpo-SAM (22.07 ± 0.81%) or Hpo-SAM (22.06 ± 0.50%) with increase in the median open-circuit voltage (*Voc*) from 1.11 V (Me-4PAcz) to 1.15 V (Epo-SAM) and median fill factor (FF) from 78.47% (Me-4PAcz) to 80.32% (Epo-SAM), probably due to the smallest steric hindrance of EPA with the best effect on voids filling. To further confirm it, we tried using Me-4PACz itself to fill the voids (Mpo-SAM) and the performance of the corresponding devices (20.99 ± 0.52%) is similar to the control device (Me-4PACz) and barely any difference is observed for all the photovoltaic parameters (Figure S4, Supporting Information), indicating that the pre-requisite for selecting the second component of po-SAM is the simple molecule structure with small steric hindrance.[32,33]

To directly show the improvement in SAM coverage by po-SAM strategy, Photo-induced Force Microscopy (PiFM) measurements were conducted for the samples of FTO/Me-4PACz, FTO/Epo-SAM, FTO/Bpo-SAM and FTO/Hpo-SAM, excited by a pulsed quantum cascade laser with a wavenumber of 1234 cm^{-1}, which is the featured stretching vibration of P = O bond in Me-4PACz or the alkylphosphonic acids,[34] detected by Fourier transform infrared spectroscopy (Figure S5, Supporting Information). As shown in Figure 1b–e, the photo-induced surface potentials for each sample are 290 ± 231, 185 ± 93,

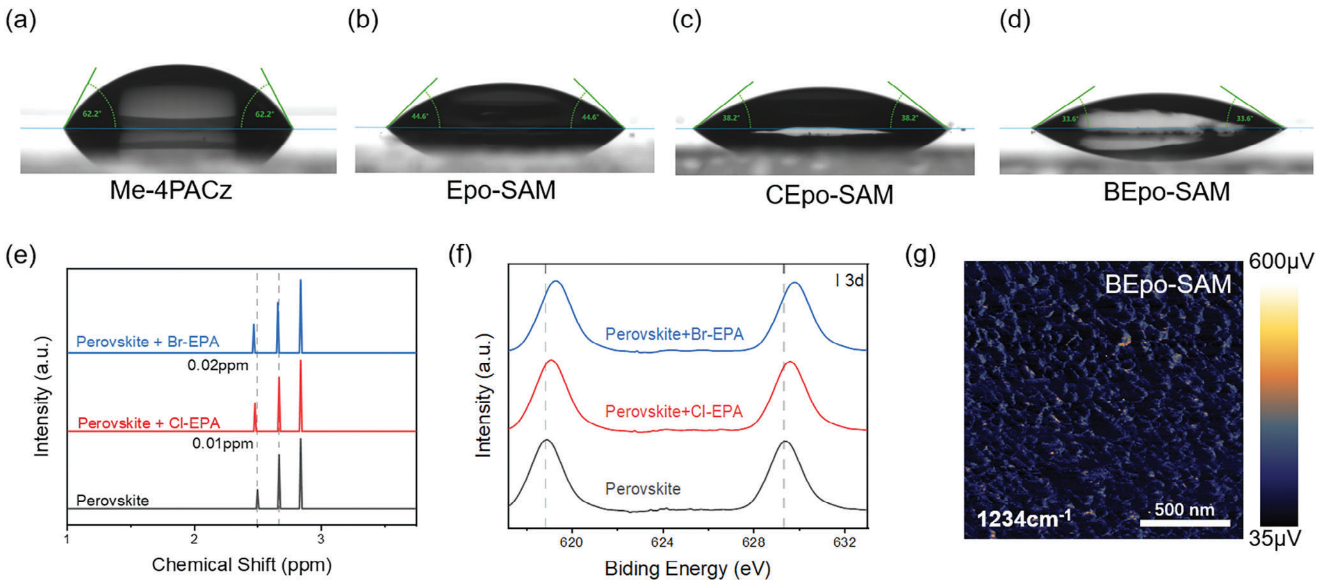

Figure 2. Perovskite contact angle tests of a) FTO/Me-4PACz, b) FTO/Epo-SAM, c) FTO/CEpo-SAM, and d) FTO/BEpo-SAM. e) ^{1}H NMR spectroscopy of perovskite (black line), perovskite mixture with Cl-EPA (red line) and Br-EPA (blue line) dissolved in (Methyl sulfoxide)-d6. f) XPS of the I 3d core level of perovskite (black line), perovskite mixture with Cl-EPA (red line) and Br-EPA (blue line), respectively. g) Photo-induced surface potential mapping at wavenumber of 1234 cm^{-1} for FTO/BEpo-SAM. The solvent of the perovskite solution for the contact angle tests is a mixture of DMF and DMSO at a volume ratio of 4:1.

266 ± 197, and 241 ± 218 μV, respectively, demonstrating that the post-assembled alkylphosphonic acids can fill the voids of SAM with decreased absolute value and narrower distribution of the surface potential.[10,26] Among the three po-SAMs, Epo-SAM shows the lowest median of surface potential with the best uniformity, which is consistent with the photovoltaic performance of the devices in Figure S4 (Supporting Information). We also performed a scanning electron microscope with energy dispersive spectroscopy (SEM-EDS) for the four samples (Figure S6, Supporting Information) to show the mapping of phosphorus in SAMs and the tendency is consistent with PiFM. By counting the phosphorus signals in EDS mapping, the weight ratio of phosphorus in the po-SAMs is increased by 13% (Epo-SAM), 10% (Bpo-SAM), and 6.5% (Hpo-SAM), respectively, compared with Me-4PACz (Figure S7 and Table S1, Supporting Information).

As the introduction of alkylphosphonic acids may affect the properties of the interface, the static contact angles with perovskite were measured (**Figure** 2a,b; Figure S8a–d, Supporting Information) and are 62.2°, 44.6°, 50.4°, and 60.5°, respectively, for the samples of FTO/Me-4PACz, FTO/Epo-SAM, FTO/Bpo-SAM, and FTO/Hpo-SAM. To further confirm that the improved wettability comes from the alkylphosphonic acids, the static contact angle with water for FTO/EPA, FTO/BPA, and FTO/HPA were measured and are 41.1°, 45.9°, 49.8°, respectively, exhibiting the same tendency with the po-SAM samples (Figure S8e–g, Supporting Information). Therefore, the best wettability of Epo-SAM should be ascribed to EPA with the shortest hydrophobic alkyl chain. To further improve the wettability of po-SAM for perovskite and realize the defects passivation at the buried interface of perovskite, the head group of EPA can be substituted with halogen such as Cl, Br, I to increase the polarity of the SAM surface and form halogen bonding with the perovskite.[35–37] In this work, we mainly focus on the discussion of Cl-EPA and Br-EPA because the raw material of PI_3 for synthesizing I-EPA is unstable. In Figure 2b–d, compared with FTO/Epo-SAM, the static contact angles with perovskite are lowered from 44.6° to 38.2° and 33.6° for the po-SAMs formed by Me-4PACz and Cl-EPA (CEpo-SAM) and Me-4PACz and Br-EPA (BEpo-SAM), respectively, due to the stronger chemical interaction between the halogen and the perovskite precursor solution. Benefiting from the improvement in the wettability, the perovskite film could easily realize the full coverage of the substrate, as shown in Figure S9 (Supporting Information).[38,39] Accordingly, the PSCs based on the CEpo-SAM and BEpo-SAM showed improved reproducibility compared with the Me-4PACz based devices and for each batch, twelve devices were fabricated (Figure S10, Supporting Information). In addition, the median PCE is improved from 21.24 ± 0.62% to 22.88 ± 0.39% (CEpo-SAM) and 23.76 ± 0.35% (BEpo-SAM), respectively, probably due to the formation of halogen bonding between halogen-EPA and the undercoordinated I^- at the buried interface of the perovskite, which could have an effect on the perovskite crystallization and defects passivation.[40]

The coordination between halogen-substituted EPA and the perovskite were characterized by the nuclear magnetic resonance (NMR) spectroscopy and X-ray photoelectron spectroscopy (XPS). Samples in NMR test were fabricated by mixing perovskite with Cl-EPA or Br-EPA in (Methyl sulfoxide)-d6. Compared with the pristine perovskite, the chemical shifts of H in the perovskite mixed with Br-EPA (0.02 ppm) are larger than that of the H in the perovskite mixed with Cl-EPA (0.01 ppm) (Figure 2e), indicating stronger coordination between Br-EPA and perovskite than that between Cl-EPA and perovskite. We further conduct XPS characterization in Figure 2f. In the XPS of I 3d core level, there are two main peaks of I 3d 3/2 (629.34 eV) and I 3d 5/2 (618.81 eV) in the perovskite, which shifted to 629.61 and 619.12 eV with the addition of Cl-EPA, while the two peaks shifted to the higher binding energy of 629.77 and 619.31 eV with the addition of Br-EPA, indicating more electrons transfer from the perovskite to Br-EPA than Cl-EPA.[41] The NMR and XPS results confirm the stronger halogen bonding between Br-EPA and the undercoordinated I^- in perovskite than that for Cl-EPA.

Before the investigation of the effect of halogen bonding on the perovskite crystallization and defects passivation, we need to confirm the coverage of the new BEpo-SAM. The PiFM measurement was performed for FTO/BEpo-SAM and the conditions are the same as in Figure 1b–e, which exhibits a low photo-induced surface potential of 117 ± 56 μV (Figure 2g). The much lower median photo-induced surface potential than the samples in Figure 1b–e should be ascribed to the superposition of the C-Br characteristic octave peak (ν C-Br 601 cm^{-1}, Figure S11, Supporting Information) and the P═O response peak (Figure S5, Supporting Information). Fortunately, the uniformity of BEpo-SAM can still be demonstrated by the narrow amplitude. To further confirm the uniformity of BEpo-SAM, the SEM-EDS result is shown in Figure S12 (Supporting Information) and the weight ratio of phosphorus in BEpo-SAM is increased by 12% compared with Me-4PACz, which is close to the SEM-EDS result of Epo-SAM in Figure S6b (Supporting Information).

The morphologies of the perovskites grown on Me-4PACz and BEpo-SAM were characterized by atomic force microscopy (AFM) and SEM. The height images of the perovskite and the corresponding AFM section analysis profiles were shown in Figure S13 (Supporting Information). The root mean square (RMS) roughness of the perovskite is reduced from 14.55 to 8.75 nm by applying BEpo-SAM as the substrate, indicating smoother perovskite surface with less nonradiative recombination centers such as the dangling bonds and vacancies. In addition, the top-view SEM images show the enlarged grain size with less grain boundaries and narrower grain size distribution for the perovskite deposited on BEpo-SAM (Figure S14, Supporting Information). The structural properties of the perovskite films were further characterized by X-ray diffraction (XRD) in Figure S15 (Supporting Information). The perovskite films grown on BEpo-SAM show stronger (110) crystallographic diffraction and the full width at half maximum (FWHM) is obtained by fitting the characteristic peaks of (110). Compared with the perovskite on Me-4PACz, the FWHM is reduced from 0.31° to 0.25° for the perovskite on BEpo-SAM, probably due to the formation of highly directional halogen bonding between Br-EPA and perovskite.[42]

The carrier dynamics for the perovskite films deposited on Me-4PACz, Epo-SAM, and BEpo-SAM are investigated by the steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) spectra (Figures S16 and S17, Supporting Information), and the detailed parameters for double exponential fitting of TRPL are shown in Table S2, (Supporting Information). Compared with the perovskite film on Me-4PACz, stronger PL intensity and prolonged carrier lifetime from 488 to 585 ns are observed for the perovskite film on Epo-SAM, which should be ascribed to the voids-filling effect of EPA. As for the perovskite film on BEpo-SAM, the PL intensity and the carrier lifetime (790 ns) are further enhanced due to the improved crystallization and defects passivation effect of halogen bonding.[43–45]

To probe the defect nature in the devices, the space charge limited current (SCLC) method that analyses the photovoltaic performance (*I*–*V*) of electron- or hole-only devices is a simple way, which can be usually divided into three regions: the Ohmic region with a linear increase in current and voltage in the low voltage range, a nonlinear polynomial increase within the range of the trap-filling limited (TFL) region at medium voltage and the trap position at the inflection point completely filled by charge carriers. The intermediate voltage range between the Ohmic region and the TFL region is called the trap-filling limited voltage (V_{TFL}), and the trap density can be determined by the following equation:[46]

$$V_{\mathrm{TFL}} = \frac{e n_{\mathrm{t}} L^2}{2\varepsilon_{\mathrm{r}}\varepsilon_0} \tag{1}$$

where ε_r and ε_0 are the relative dielectric constant of perovskite and vacuum permittivity, L is the thickness of perovskite film, and n_t is the electron (hole) trap density of perovskite. In **Figure 3a**, the V_{TFL} is deduced by the cross points of ohmic region and trap-filling region, which are 0.23 and 0.11 V for the devices based on Me-4PACz and BEpo-SAM, respectively. According to the equation, the calculated hole trap-state density is reduced from 1.13 × 10^{15} cm^{-3} (Me-4PACz) to 5.40 × 10^{14} cm^{-3} (BEpo-SAM). Additionally, the light intensity-dependent *V*oc is plotted (Figure 3b) and the fitting relationship is:[47,48]

$$V_{\mathrm{OC}} = nkT \ln L/q \tag{2}$$

where n, k, T, L, and q represent the ideality factor, Boltzmann constant, absolute temperature, light intensity, and elementary charge, respectively. The device with BEpo-SAM shows the smaller ideality factor of 1.38 than that of the device with Me-4PACz (1.52), indicating suppressed Shockley–Read–Hall recombination at the interface because of the voids filling and defects passivation. From the electrochemical impedance spectroscopy (EIS) in Figure 3c, it is calculated that the device with BEpo-SAM has the higher recombination resistance (22 624 vs 36 472 Ω) and smaller series resistance (19.39 vs 11.25 Ω), which demonstrated less current leakage and efficient charge transport than the devices based on Me-4PACz. The charge extraction was characterized by transient photocurrent decay (Figure 3d) and the photocurrent decay decreased from 5.28 μs (Me-4PACz) to 3.10 μs (BEpo-SAM). The above results demonstrated the reduced nonradiative charge recombination and more efficient charge extraction and transport in the devices based on BEpo-SAM.

Finally, we modified the PSCs by applying top-surface perovskite passivation with phenethylammonium chloride and the anti-reflection layer of MgF_2. The cross-sectional SEM image of the device based on BEpo-SAM is shown in **Figure 4a** and the detailed fabrication process for each layer and the structure is illustrated in the supporting information and Figure S18 (Supporting Information). The *I*–*V* curves for the devices with different substrates are shown in Figure 4b and the device using Me-4PACz as the hole-selective layer obtained a PCE of 22.87% with the *V*oc, short-circuit current density (*J*sc) and FF of 1.13 V, 24.92 mA cm^{-2} and 81.21% under reverse scan, respectively. In comparison, the device with an optimal concentration of 0.2 mg ml^{-1} (Figure S19 and Table S3, Supporting Information) obtained a PCE of 25.16% with the *V*oc, *J*sc and FF of 1.20 V, 25.10 mA cm^{-2} and 83.54%, respectively, under reverse scan. The significant improvement in *V*oc and FF and reduced hysteresis factor *H*i from

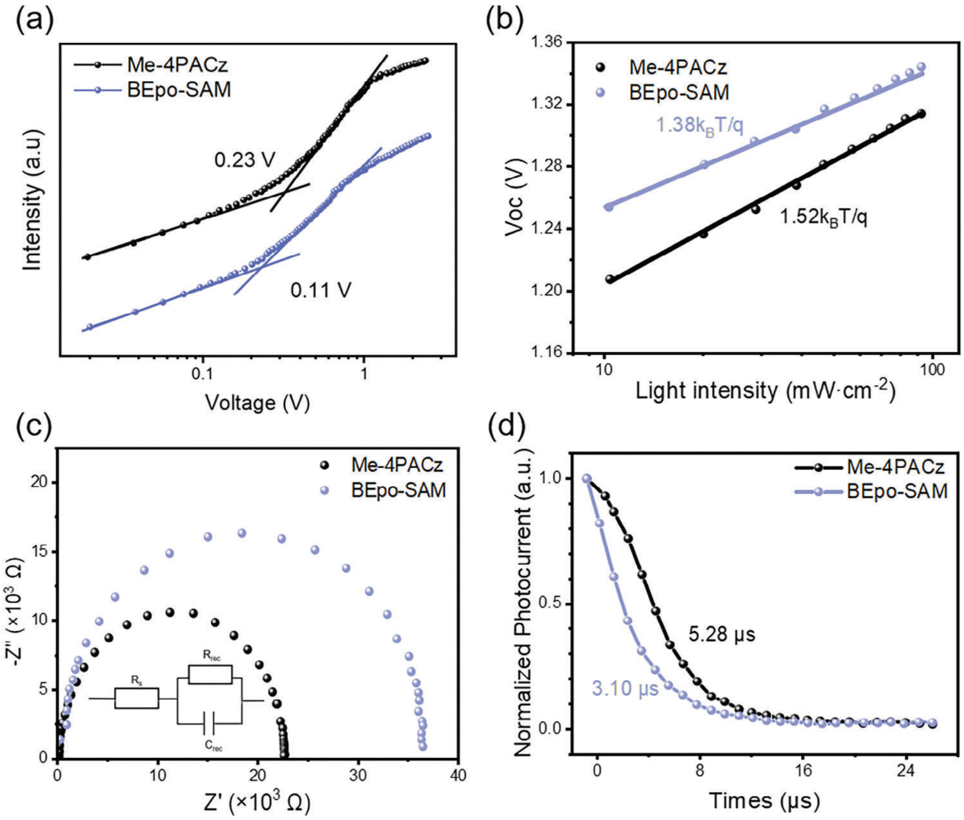

Figure 3. a) SCLC test for the devices with Me-4PACz or BEpo-SAM. b) *Voc* versus light intensity for the devices with Me-4PACz or BEpo-SAM. c) Nyquist plots of the EIS for the devices with Me-4PACz or BEpo-SAM, the inset is the equivalent circuit, where *R*s is the series resistance and *R*rec is the charge recombination resistance. d) Transient photocurrent decay curves for the devices with Me-4PACz or BEpo-SAM.

1.5% to 0.8% (Table S4, Supporting Information) can be mainly attributed to the more compact po-SAM with less voids and the passivation effect of Br-EPA at the buried interface. The external quantum efficiency (EQE) curves for the control and the target devices are displayed in Figure 4c for Jsc calibration. The integrated Jsc of the devices based on Me-4PACz and BEpo-SAM are 24.86 and 25.35 mA cm^{-2}, respectively, which match well with the observed *J*sc in *I–V* curves. The increase in the quantum efficiency can be attributed to the improved perovskite crystallization. One of the devices based on BEpo-SAM is sent to an accredited independent PV test laboratory (Shanghai Institute of Microsystem and Information Technology (SIMIT), Shanghai, China) for certification and a PCE of 24.67% under forward scan and 24.61% under reverse scan was certified (Figure S20, Supporting Information) with negligible hysteresis.

The dense SAM, the defects passivation, and the improved crystallinity of perovskite have been reported to enhance the operation stability of PSCs. Here, we verified the operation stability of our devices in Figure 4d. The encapsulated device based on BEpo-SAM maintained ca. 90% of the initial PCE (24.92%) after 1000 h of operation at the maximum power point (MPP), under 1-sun illumination and 40 °C, whereas the PCE of the device based on Me-4PACz dropped by 51% after 600 h of operation under the same conditions. To examine the storage stability of the devices, the devices were kept in a dark dry box with relative humidity of <5% and room temperature for 8 weeks (Figure 4e). For the device based on BEpo-SAM, the efficiency increased from 24.50% to 25.05%. As for the device based on Me-4PACz, an increase in the efficiency can also be found in the first 2 weeks, whereas it went through a gradual drop and maintained 81% of the initial PCEs after 8 weeks.

3. Conclusion

In conclusion, a po-SAM strategy is proposed and realized by the post-assembled Br-EPA to fill the voids of Me-4PACz, which forms a densely packed monolayer with minimal voids to reduce the current leakage of the devices. In addition, the wettability of po-SAM for the perovskite is improved due to the short alkyl chain of Br-EPA and the strong chemical interaction between Br-EPA and the perovskite precursor solution, which increases the reproducibility of the devices. The formation of the directional halogen bonding between Br-EPA and the undercoordinated I$^-$ improves the perovskite crystallinity and passivates the trap states at the buried interface, contributing to efficient charge extraction and transport. The effect of alkyl chain length and halogen bonding strength on the properties of the po-SAM/perovskite interface and devices' performance have been investigated. Based on the newly constructed BEpo-SAM, a PCE of 25.16% is obtained with a certified efficiency of 24.67% and 24.61% under forward or reverse scan, respectively. The encapsulated device retained 90% of its initial value after operating at the MPP for 1000 h under

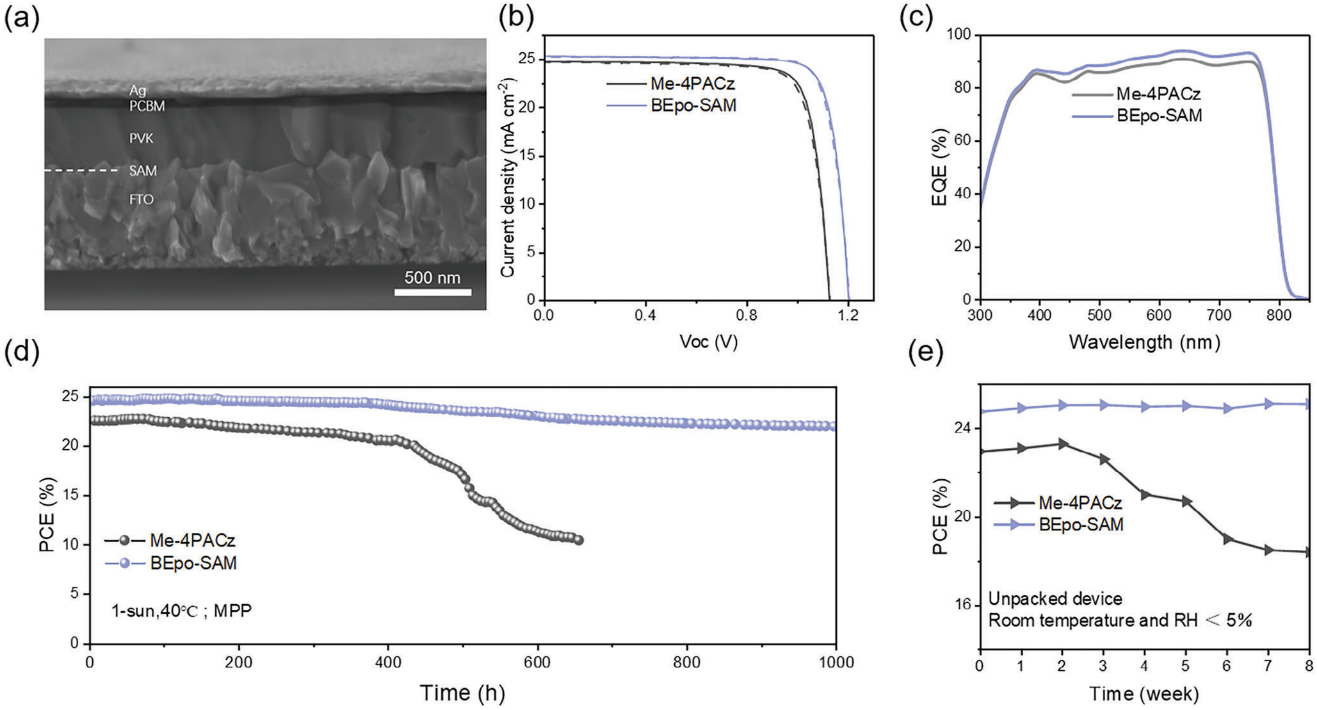

Figure 4. a) The cross-sectional SEM image of the device based on BEpo-SAM. b) *I–V* curves of the fabricated devices under reverse (solid) and forward scan (dashed). c) EQE curves of Me-4PACz and BEpo-SAM based devices. d) The operation stability of the encapsulated devices at MPP, under 1-sun illumination and 40 °C. e) The storage stability of the unpacked devices in a dry box with a relative humidity of <5% and room temperature.

continuous 1-sun illumination, 40 °C. This study shows the potential of po-SAM strategy for strengthening the buried interface of the inverted PSCs and provides a new way to modify SAM. More functionalities can be introduced through the molecule design of the po-assembled component to further advance the development of PSCs.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

alkylphosphonic acids, buried interface, inverted perovskite solar cells, post-assembly, self-assembled monolayer

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